
Predicting Chatbot Conversation Capacity

From a strong proxy to a validated final model

Course	COSC2753 Machine Learning
Assessment	Assignment 1 - Practical & Written Assignment
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Report map. Phase 0 is the notebook's execution guide. The five assessed pages then follow its eight analytical phases in order: Phases 1–2 (problem and audit), Phases 3–4 (grouped EDA and policies), Phase 5 (model competition), Phase 6 (diagnostic judgement), and Phases 7–8 (frozen evaluation, prediction and final verification). IEEE references are followed by one appendix page of additional diagnostic evidence.

Workflow note. The notebook is intentionally detailed because I used this assignment to explore a practical ML workflow. It records data roles, leakage controls, decision checks, model freezing and output-file verification. This report summarises the evidence used for each decision.

I declare that this report presents my own analysis. Published methods and software are acknowledged in the reference list; every reported experiment and result is verifiable in the submitted notebook.

Phase 1 - Problem Definition & Evaluation Contract

- **Prediction task - define “best” before modelling.** The task is supervised regression of `TARGET_Capacity`, the meaningful conversation capacity of a chatbot system [1]. I define the best model as one that generalises to unseen Countries, has low MAE and controlled RMSE, and remains interpretable about failure conditions - not simply the lowest in-sample error.
- **Generalisation unit - Country, not row.** Of 136 labelled Countries, 129 repeat and seven are singletons; development contains five singleton Countries. Later development-only EDA finds Country-mean target SD 9.10 versus median within-Country SD 2.97. A random row split could therefore place the same Country in fitting and validation data and overstate performance. Blocking structured data during validation is recommended when dependence can otherwise leak across folds [2].
- **Evaluation protocol - development decisions first, holdout after freeze.** A seeded Country-group split reserves 417 rows from 28 Countries as a locked holdout. All model design uses the remaining 1,653 rows from 108 Countries and five fixed Country-grouped folds with zero Country overlap. Every learned imputation, transformation, scaling and coefficient is fitted inside the training part of each fold.
- **Metrics - MAE as primary, RMSE for large errors.** MAE is primary because it reports typical error directly in target units. RMSE shows rare large misses, R^2 provides mean-baseline context, and signed residual bias diagnoses systematic over- or under-prediction.

Phase 2 - Data Loading, Structural Audit & Data Quality

- **Preserve information; remove only by role.** The source contains 2,071 rows and 26 columns. One row with a missing target is excluded without target imputation, leaving 2,070 labelled rows. Nine predictor cells are missing across nine rows, and one `Status=15` contradicts its documented binary domain. `RecordID` is retained only for traceability and CSV ordering; it is never a predictor [3].

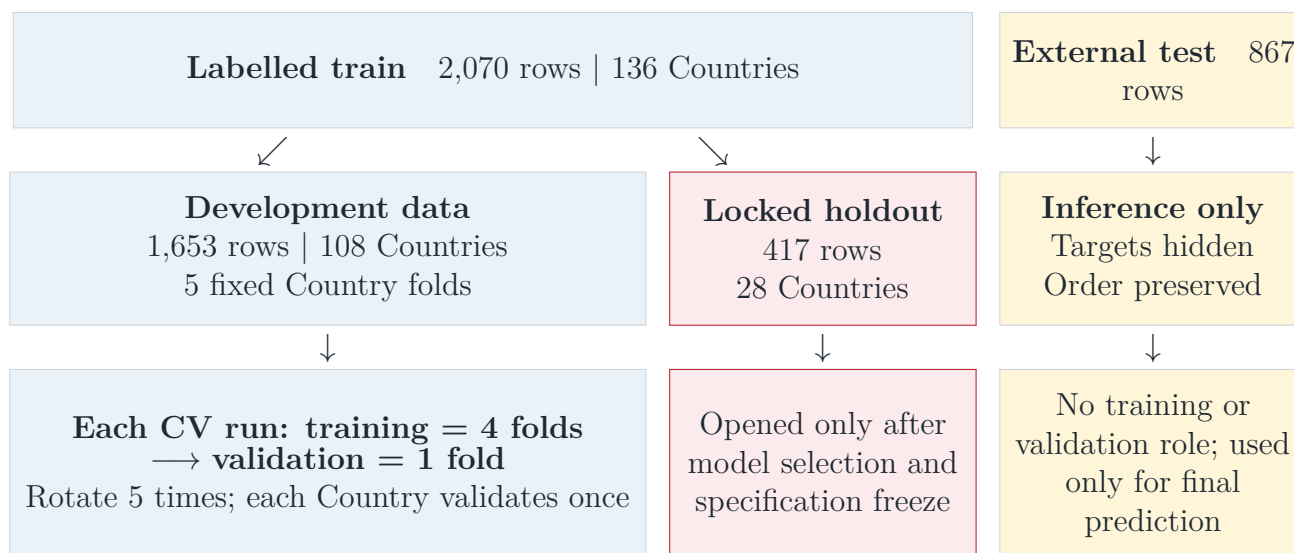


Figure 1: Phase 2 data-role map. The same development data are reused across five grouped-CV runs. In each run, four Country folds train the model and one fold validates it; the validation fold changes each run.

Phase 3 - Country-Grouped Split & Development-Only EDA

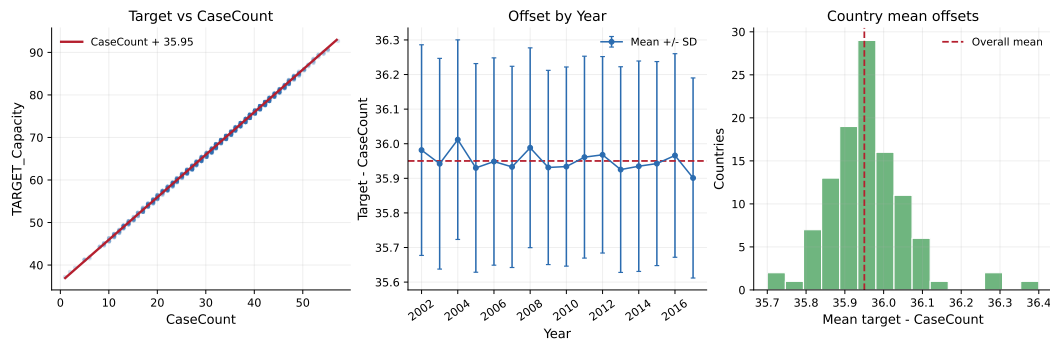


Figure 2: *Development-only evidence. The target is close to $\text{CaseCount} + 35.95$, while Year and Country summaries show a stable offset. This supports prediction but does not prove a causal relationship.*

- **Leakage boundary - explore only after the split is frozen.** All plots and relationships on this page use the 1,653-row development partition; the 417-row, 28-Country holdout remains untouched.

EDA-to-design summary. Country-mean target SD 9.10 versus median within-Country SD 2.97 supports Country-grouped evaluation. Eight lower target flags (0.48%) are retained, so MAE is paired with RMSE and residual checks. Nine missing predictor cells and one invalid `Status=15` motivate fold-fitted medians and a tested `invalid_to_nan` policy. Ten non-negative, right-skewed predictors become model-family-specific `log1p` candidates; all 23 eligible predictors remain available to multivariable models. Finally, 1,651 complete `CaseCount` pairs give Pearson 0.9995, Spearman 0.9994 and a stable mean offset of 35.9502 (SD 0.2940; range 35.5–36.4), motivating the Offset baseline and a mandatory Phase 6 proxy ablation.

Phase 4 - Preprocessing, Policy Ablations & Baselines

The EDA defines candidate actions, while the same five grouped folds decide which policies are retained. No predictor is removed from EDA alone.

Table 1: *One-factor policy ablations on the same five grouped folds. CV MAE is the average absolute prediction error: 3.8108 means predictions are off by about 3.81 target units; lower is better. Relative change compares with that reference: negative improves and positive worsens (e.g., -6.7% means 6.7% less error).*

Change from proxy-controlled OLS	CV MAE	Relative change	Decision
Reference without <code>CaseCount</code>	3.8108	–	Risk control
Country one-hot encoding	7.7234	+102.7%	Reject
Invalid Status to missing	3.7337	-2.0%	Select
Add <code>CaseCount</code>	0.2763	-92.8%	Conditional
Evidence-supported <code>log1p</code> ¹	3.5542	-6.7%	Family-specific

Phase 4 decision. Freeze `invalid_to_nan`, use Country as `group_only`, and retain `CaseCount` conditionally; Phase 6 must test whether accuracy survives without the proxy.

¹The ten features whose absolute skewness decreased after `log1p` are `InferenceLatency`, `AIInfraExpPercent`, `SecurityBreachIncidents`, `UptimeReliabilityIndex`, `SystemVulnerabilityScore`, `GlobalAdoptionIndex`, `ActiveUserBase`, `TeenQueryEfficiency`, `ChildQueryEfficiency`, and `Complexity`.

Phase 5 - Candidate Comparison, Tuning & Finalist Freeze

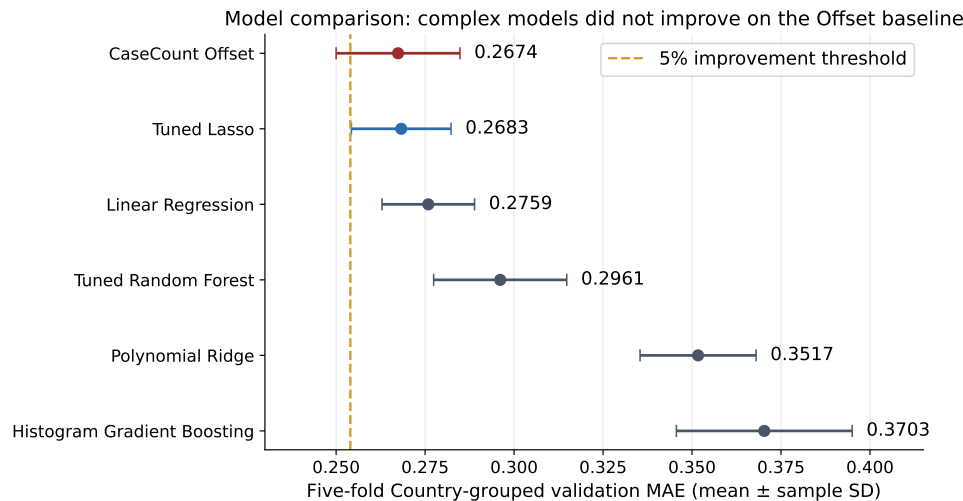


Figure 3: Shared Country-grouped evidence. Error bars are sample SD across the five fixed folds. The predefined 5% line requires a visible MAE reduction before accepting extra model complexity.

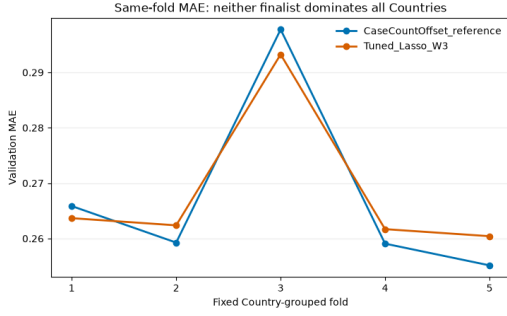
- **Models compared - linear, regularised, tree-based and boosting.** I compare Week 2 linear regression; Week 3 Polynomial Ridge and Lasso [4]; a Week 5 Random Forest extension [5]; and beyond-class histogram gradient boosting [6]. All use identical rows, folds, metrics and fold-local pipelines implemented in scikit-learn [7].
- **Initial result - more complex models did not improve validation.** Lasso is the best multivariable model (0.2689 ± 0.0147) but trails Offset (0.2674 ± 0.0174), so it is the only linear model tuned. Random Forest and boosting score $0.2974/0.3703$, with gaps $0.1662/0.2327$; only the forest is tuned because depth and leaf controls target its gap. Weaker Ridge and boosting are not taken forward.
- **Tuning tests model-specific capacity controls.** Standardised Lasso uses the bounded log-spaced grid $\alpha \in \{0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1\}$ around the initial 0.01; lower and higher values test weaker shrinkage versus added sparsity. Thirty-five CV fits select $\alpha = 0.03$, moving MAE only to 0.2683 ± 0.0140 . The forest's 0.1662 train-validation gap motivates the bounded grid depth $\{\text{None}, 8\} \times \text{leaf} \{2, 8\} \times \text{feature fraction} \{0.5, 1.0\}$, with 400 trees fixed. Forty CV fits select $(8, 2, 1.0)$, but MAE remains 0.2961 ± 0.0187 and the gap 0.1276. Both searches reuse the fixed grouped folds with the holdout closed; tuned scores are selection evidence, not independent nested-CV estimates.

Table 2: Grouped-CV results used for model selection.

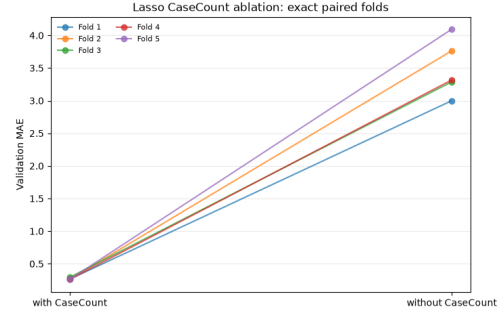
Model	Course role	CV MAE	Train-CV gap	Outcome
CaseCount Offset	Specific reference	0.2674 ± 0.0174	-0.0008	Finalist
Linear regression	Week 2	0.2759 ± 0.0130	0.0065	Reference
Polynomial Ridge	Week 3	0.3517 ± 0.0163	0.0961	Reject
Tuned Lasso	Week 3	0.2683 ± 0.0140	0.0017	Finalist
Tuned Random Forest	Week 5	0.2961 ± 0.0187	0.1276	Reject
Histogram GB	Additional	0.3703 ± 0.0247	0.2327	Reject

Finalist selection. Offset has lower mean MAE in three folds; Lasso in two. Their mean difference is only 0.0009 (0.32% of Offset error), so both proceed to paired error, robustness and interpretation checks.

Phase 6 - Diagnostics, Robustness & Ultimate Judgement



(a) Same-fold finalist MAE.



(b) Lasso with/without CaseCount.

Figure 4: The finalists trade fold wins. When CaseCount is removed, Lasso MAE increases in every fold, so it is not an independent fallback model.

- **Near tie in validation performance.** Exact-fold OOF reconstruction gives one prediction per finalist for every development row. Offset and Lasso pooled MAE are 0.2676 and 0.2684; Lasso has lower absolute error on 835 rows and Offset on 818. The changing winner across folds and rows does not justify a claim of statistical dominance.
- **Shared failure - accuracy depends on proxy availability.** Both missing-CaseCount rows appear in both top-20 failure lists. The largest error row has actual target 55.6 but predictions near 70.9, creating an error about 15.35. The two-row missing-CaseCount slice has MAE 8.20; $n = 2$ makes its frequency unstable, but does not negate the observed 15.35-unit magnitude.
- **Ablation - Lasso is not a reliable fallback.** Removing only CaseCount raises tuned-Lasso MAE from 0.2683 ± 0.0140 to 3.4918 ± 0.4334 (+1,201.56%); RMSE rises from 0.4197 to 4.5469 and every fold degrades. Seven of 23 coefficients are non-zero, but the standardised CaseCount coefficient is 9.4895 while every other full-refit magnitude is at most 0.0369.
- **Simplicity - show the CaseCount dependence.** Offset has lower mean MAE/RMSE, near-zero OOF bias and a fold-stable fitted offset (SD 0.0063); fold-fitted median imputation is explicit. Lasso provides neither the required 5% improvement nor lower dependence on CaseCount.

Table 3: Final comparison before holdout evaluation.

Criterion	Offset	Tuned Lasso	Reading
CV MAE	0.2674 ± 0.0174	0.2683 ± 0.0140	Practical near-tie
CV RMSE	0.4143	0.4197	Offset slightly lower
Fold wins	3/5	2/5	No consistent dominance
Proxy dependence	Explicit	+1,201.56% ablated	Both high-risk
		MAE	
Complexity	1-feature explicit rule	23-feature pipeline	Offset easier to audit

Ultimate judgement - select CaseCountOffset_reference. The small numerical difference does not decide the choice. Lasso neither meets the required 5% improvement nor reduces dependence on CaseCount. Offset shows the same dependence with one feature, a fixed slope, a fitted offset and a median-based fallback. This choice does not claim causality, fairness, the best possible model or production readiness.

Phase 7 - Frozen Holdout & Prediction Submission

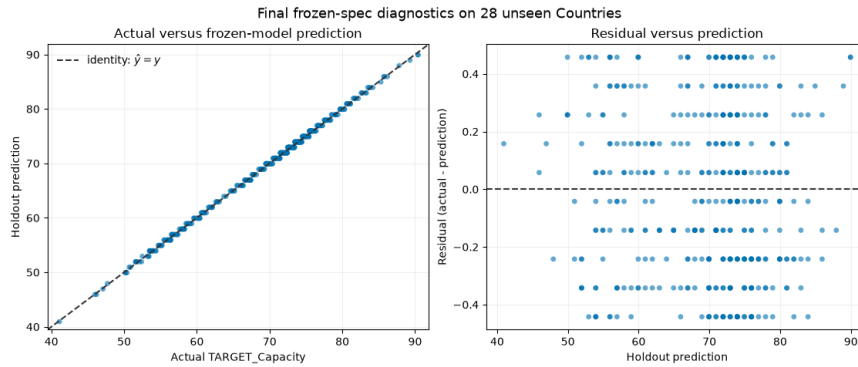


Figure 5: Frozen-spec evaluation on 417 rows from 28 unseen Countries. The holdout has no missing CaseCount, so it cannot test the missing-proxy failure.

Table 4: Frozen evidence. CV MAE/RMSE/ R^2 are fold means; CV bias is pooled OOF.

Evidence	MAE	RMSE	R^2	Bias
Five-fold Country CV	0.2674	0.4143	0.9971	-0.0008
Locked 28-Country holdout	0.2527	0.2893	0.9990	-0.0112

- **Acceptance - consistent unseen-Country evidence.** Holdout MAE is 5.49% below CV and bias 0.0112 is within the frozen 0.0425 limit; both gates pass, versus `DummyMedian_context` MAE 6.9811. As disclosed in the notebook, a read-only audit reproduced these already-frozen metrics before the saved clean run and changed no decision.
- **Final refit - one transparent rule.** After acceptance, the unchanged specification is refitted on all 2,070 labelled rows:

$$\hat{y} = \text{CaseCount} + \hat{b}, \quad \hat{b} = 35.9399.$$

This fitted rule is the only model used for external prediction.

- **Three working assumptions define when the model can be used.** CaseCount must be available pre-prediction and not target-derived; its measurement must remain consistent; and the additive relationship must continue for new Countries. The CSVs cannot prove these assumptions.
- **External range check - report extrapolation without changing the model.** Labelled CaseCount spans 1–57; external values span -5–61 with no missing values. Three negative task counts are unexpected for a count under the metadata and below support; six values are above support [1]. Because this is a post-freeze audit, all nine extrapolations are disclosed but neither corrected nor clipped.

Phase 8 - Final Code & Prediction CSV Verification

The gate verifies 867 finite predictions, exact `ID,TARGET_Capacity` columns, unique IDs and unchanged template order; it performs no refit, retuning or prediction change.

Final recommendation. Use the frozen Offset model for the assignment prediction file, conditional on the three CaseCount assumptions. If CaseCount is unavailable, changes meaning, or no longer follows the target, the predictions may be unreliable. Production use would require feature monitoring and tests for missing CaseCount.

References

- [1] RMIT University, “COSC2753 Assignment 1 dataset metadata,” course material, 2026.
- [2] D. R. Roberts et al., “Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure,” *Ecography*, vol. 40, no. 8, pp. 913–929, 2017, doi: [10.1111/ecog.02881](https://doi.org/10.1111/ecog.02881).
- [3] RMIT University, “COSC2753 Machine Learning - Assignment 1, updated version,” assignment specification, 2026.
- [4] R. Tibshirani, “Regression shrinkage and selection via the Lasso,” *J. Roy. Stat. Soc. B*, vol. 58, no. 1, pp. 267–288, 1996, doi: [10.1111/j.2517-6161.1996.tb02080.x](https://doi.org/10.1111/j.2517-6161.1996.tb02080.x).
- [5] L. Breiman, “Random forests,” *Mach. Learn.*, vol. 45, pp. 5–32, 2001, doi: [10.1023/A:1010933404324](https://doi.org/10.1023/A:1010933404324).
- [6] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, 2001, doi: [10.1214/aos/1013203451](https://doi.org/10.1214/aos/1013203451).
- [7] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: <https://www.jmlr.org/papers/v12/pedregosa11a.html>

How the references support the report

Table 5: *Traceability from published sources to report decisions.*

Decision area	Sources	Use in this report
Dataset metadata	[1]	Define feature meaning and expected domains, including why negative CaseCount values require a warning.
Validation design	[2]	Support Country blocking so repeated rows from one Country cannot cross fitting and validation data.
Assignment rules	[3]	Define the target, ID role and prediction-file contract.
Regularised linear model	[4]	Define the Lasso model used as the main multivariable finalist.
Random Forest	[5]	Establish the bagged tree model compared in Phase 5.
Gradient boosting	[6]	Establish the boosting model family compared in Phase 5.
Implementation	[7]	Identify the software used for preprocessing, grouped validation, tuning and final fitting.

Source boundary. Course materials define the task, data and submission rules. Published papers support validation and model choices; they do not supply the reported scores. Every numerical result comes from the submitted data and notebook.

Appendix - Additional Diagnostic Evidence

The main report contains the evidence used for model selection. This appendix adds two checks that help define when the selected model may fail.

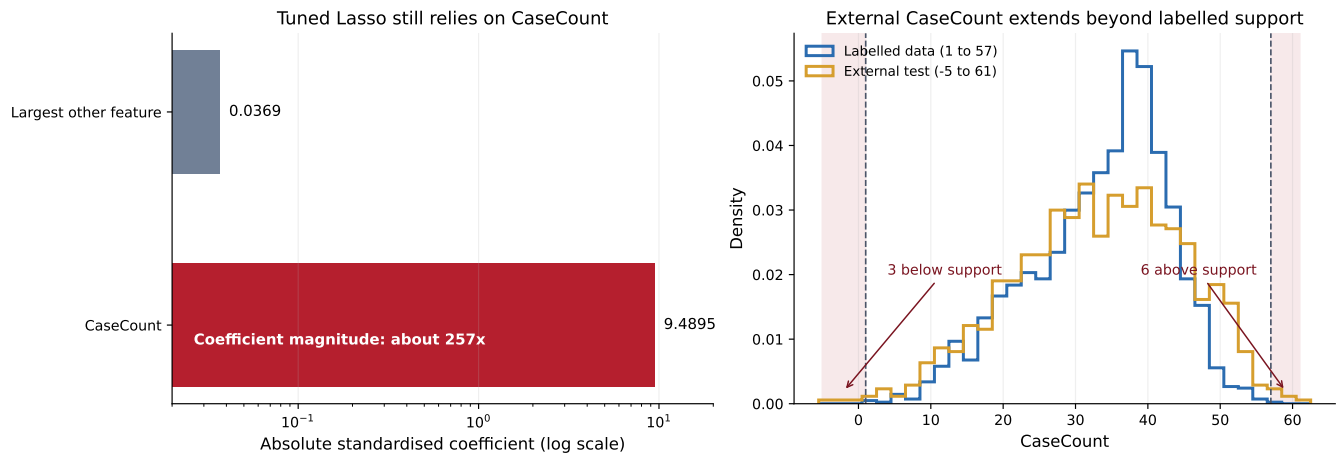


Figure A1: Additional CaseCount checks. Left: the fitted Lasso coefficient for CaseCount is about 257 times the largest other coefficient, so Lasso is not an independent fallback. Right: nine external rows are outside the labelled CaseCount range. Their targets are unknown, so this is a warning rather than evidence of prediction error.

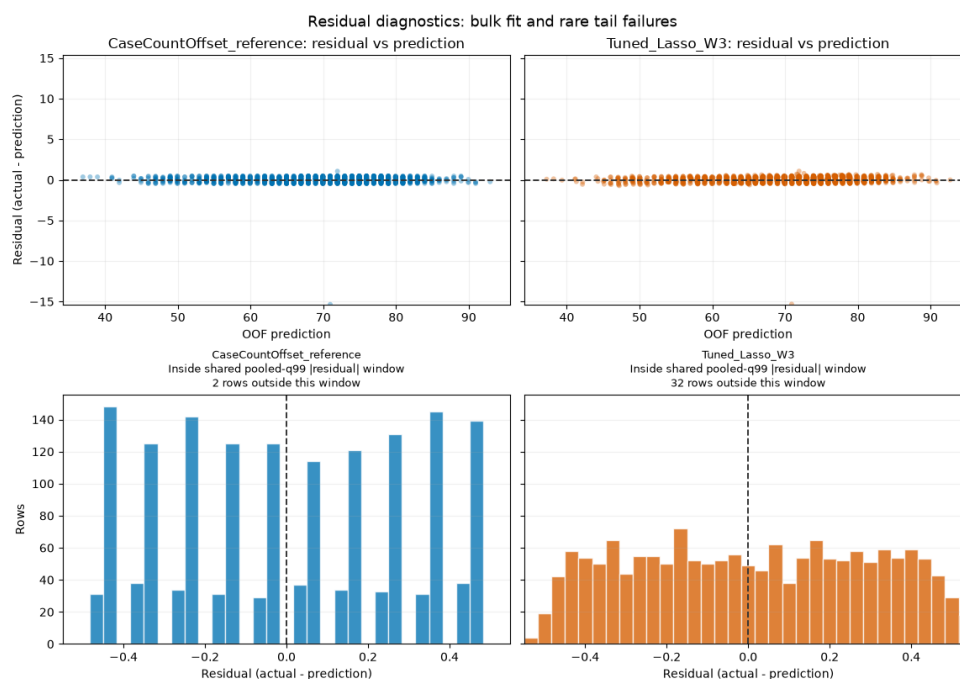


Figure A2: Exact-fold OOF residual checks for the selected Offset model. Most errors are small, but the missing-CaseCount rows form a clear error tail. This supports the stated requirement that CaseCount must be available at prediction time.

Action boundary. Require CaseCount before prediction; flag the nine external rows outside labelled support without clipping them; and revalidate the model if CaseCount meaning or its target relationship changes.